HENRY DEUTSCH

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Portfolio: **henry-deutsch.com**

WORK EXPERIENCE

Tenex | *Forward Deployed Engineer — Full Time* Jul 2026 - Nov 2026

- Replaced ad-hoc AI-built apps at an $87B PE firm with one governed intake path that provisioned 9 apps in its first month. An LLM judge triages requests, and each generated repo gets only the capabilities it needs, with private-network data.
- Shipped the system that reconciles $1B+ a year in capital calls and distributions across 20 fund entities. Designed it through user interviews and usability tests; it cut the fund's period reconciliation from 2 days to 10 minutes.
- Built HeySherpa, an AI nutrition coach for Dan Go (5M followers) with a 6,900-person waitlist. Designed its agent tools for meal logging, meal plans, and grocery lists, gave it per-user memory, and wrote evals that gate prompt changes

NewForm AI | *Software Engineer I — Full Time* Aug 2025 - Mar 2026

- Built WillBot's LLM agent harness for an ad-analytics platform serving 90+ consumer brands, enabling natural-language analysis across $100M+ in paid media spend. Added regression tests for single- and multi-client queries.
- Led major refactor making the platform ad-network-agnostic, unlocking $50K+/mo in TikTok spend. Refactored 31K+ LOC across 140+ tRPC endpoints and rebuilt the Bayesian inference engine around platform-agnostic objects.
- Designed and deployed a high-throughput ad similarity detection service using perceptual hashing, authoring a white paper and processing 359K ads across 20 clients to enable scalable similarity queries.

KnoWhiz | *Software Engineering Intern* May 2024 - Sep 2024

- Rearchitected Quizlet imports for an AI learning platform with 4.3K MAUs, cutting large-import failures from ~50% to <4% and saving 200 user hours/month. Used a two-phase commit pattern with WebSocket progress updates.
- Mentored developer while owning the import rearchitecture, reviewing 40+ of their PRs through to merge.

HF Engineering | *Software Engineering Intern* May 2024 - Sep 2024

- Led development of a Next.js onboarding and project-management platform for 18,000 open-source volunteers.
- Integrated AWS Amplify for CI/CD; coordinated Azure DevOps releases with feature flags and rollbacks across 8 teams.

Johns Hopkins Research | *Software Development Intern* May 2023 - Aug 2023

- Designed a Go microservice for NIST's disaster-relief platform, ingesting sensor data at 1K msg/sec with 250ms latency.
- Integrated Envoy with Istio and Prometheus to add tracing and telemetry across REST APIs and microservices.

EDUCATION & CERTIFICATIONS

Johns Hopkins University | Bachelor of Science in Computer Science | Baltimore, MD Aug 2021 - May 2025

Graduated with Departmental Honors in Computer Science

Courses: Full-Stack JavaScript, Artificial Intelligence, Machine Learning, Practical Generative AI, Computer Graphics, Intermediate Programming (C++), Computer System Fundamentals, Data Structures, Algorithms, Calculus III

Awards: Placed 3rd of 43 teams, won $250 prize at HopHacks Hackathon for reimagining grocery checkout with computer vision

Certification: Claude Certified Architect, Foundations | Anthropic (Oct 2026)

SOFTWARE PROJECTS

Chief of NYC Fire Dept Website | Paid Contract — Full Stack Developer Sep 2021 - Sep 2022

- Remade website, increasing book sales $16K YoY (3x) for client; site traffic went from <100 to >3.6k visits/mo.
- Developed learning platform with React and JavaScript with quizzes, content management, and progress tracking.

N-Body Orbit Simulations | Research at Johns Hopkins Aug 2023 - May 2024

- Built convolutional autoencoder with PyTorch and NumPy. Achieved 200:1 compression with .02 MSE loss on orbit data.
- Classified orbit paths by their latent space in the NN with K-means, Hierarchical, and Agglomerative clustering in Python.
- Deployed ML infrastructure on AWS EC2 and GCP Compute Engine for parallel processing of 10k+ orbit simulations.

TECHNICAL SKILLS

Full-Stack Dev: React, Next.js, React Native, Node.js, Fastify, Express, tRPC, PostgreSQL, Redis, Prisma, Tailwind
Languages: TypeScript, JavaScript, Python, Java, Go, SQL, C++, HTML/CSS
Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Bicep (IaC), Prometheus, Linux, Git
AI/LLM: LLM agents, tool calling, evals, prompt engineering, Claude Code, Anthropic and OpenAI APIs, PyTorch